

Business Requirements Document (BRD)

Interactive Grand Family Tree Platform — Crowdsourced Pedigree System

Version: 1.0.0 **Date:** July 2026 **Status:** Approved for Implementation **Target Domain:** Web Application

1. Project Overview & Objectives

The objective of this project is to build an interactive, web-based platform that enables extended family members to enter their genealogical data using a **crowdsourced model**. Each user adds their direct family lineage, while an intelligent **deduplication engine** and a **verification & review pipeline (Approval Pipeline)** merge disparate branch nodes to construct a unified, highly accurate grand family tree.

2. Stakeholders & User Roles

Role Code	Role Name	Permissions & Responsibilities
USR	End User	Create account, claim personal profile node, add/edit direct lineage, submit relation link requests, and explore the global tree.
REV	Branch Steward (Reviewer)	Review and approve pending (PENDING) relations for their lineage branch, resolve deduplication merge requests, and verify profile claim requests.
ADM	System Administrator	Full system oversight, assign branch stewards, manage system-wide settings, and access audit logs.

3. Functional Requirements

3.1 User Account & Profile Claiming

- **[FR-1.1] Account Registration:** Users can register using email or phone number with secure authentication.
- **[FR-1.2] Profile Claiming ("This is Me"):** During onboarding, users can search for their name in the existing global tree and submit a claim request.
- **[FR-1.3] Claim Verification:** Profile claim requests are routed to the designated Branch Steward for verification before granting direct node editing rights.

3.2 Infinite Canvas Interactive UI

- **[FR-2.1] Infinite Canvas Navigation:** Render tree nodes on an interactive, zoomable, and draggable canvas (Pan & Zoom).
- **[FR-2.2] Automatic Centering:** Auto-focus viewport on the logged-in user's node upon authentication.
- **[FR-2.3] Visual Status Distinction:** Distinct card styles for living individuals (vibrant colors + current age), deceased individuals (muted warm tones + life span), and pending data (dashed border + clock badge).

3.3 Interactive Data Entry via (+) Action Buttons

- **[FR-3.1] Directional Action Buttons:** Clicking any person node displays 4 contextual (+) buttons: Top (Parent), Bottom (Child), Right/Left (Spouse).
- **[FR-3.2] Quick Relation Form:** Clicking (+) opens a modal restricted to direct relationships only, auto-setting gender based on selected relation type.

3.4 Real-time Deduplication & Merge Pipeline

- **[FR-4.1] Debounced Background Search:** As the user types first name, family name, and birth year, an automated background query checks for duplicates.
- **[FR-4.2] Real-time Match Banner:** If matching score $\geq 75\%$, a banner offers options to "Link to Existing Person" or "Create New Person".
- **[FR-4.3] Side-by-Side Review Tool:** Branch Stewards receive a side-by-side comparison UI to merge duplicate records into a single node.

3.5 Approval & Verification Pipeline

- **[FR-5.1] Pending State:** Any relation added by non-reviewers is saved with `status = 'PENDING'`.
- **[FR-5.2] Public Tree Visibility:** Only relations with `status = 'VERIFIED'` are rendered in the public global family tree.

3.6 Derived Kinship Engine

- **[FR-6.1] Restriction on Derived Types:** Disallow manual entry of derived relations (e.g., Brother, Uncle, Cousin).
- **[FR-6.2] Automated Derivation:** Infer siblings (shared parents), uncles/aunts (parent's siblings), and cousins via automated graph queries over verified direct relationships.

4. Non-Functional Requirements

- **[NFR-1] Performance:** Tree extraction queries must respond in under **1.5 seconds** for viewports rendering up to 10,000 nodes.
- **[NFR-2] Data Integrity (ACID):** Strict transactional guarantees to prevent orphaned nodes during concurrent merge operations.
- **[NFR-3] Privacy Controls:** Contact details and sensitive records of living individuals are masked for non-immediate relatives.

- **[NFR-4] Scalability:** Database and architecture designed to scale seamlessly to **1,000,000+** nodes.